

Product Requirements Document (FINAL)

Product Name

Livestock Health Intelligence System (LHIS)

1. Objective

Build a digital system that enables farmers to:

- Track livestock health daily
 - Detect early risk signals before visible illness
 - Understand causes of risk
 - Take proactive action using simulation
-

2. Problem Statement

Livestock health management is currently reactive:

- Farmers detect issues only after symptoms appear
- No continuous tracking of animal health
- Data is fragmented (milk, feed, behavior tracked separately or not at all)
- No personalized insights per animal

This leads to:

- Reduced milk productivity
 - Delayed treatment
 - Financial losses
-

3. Solution

LHIS provides:

- Continuous daily tracking

- Multi-signal data fusion
 - Context-aware risk detection
 - Actionable insights
 - Simulation for decision-making
-

4. Key Innovation

Context-Aware Risk Engine

System adapts based on animal state:

State:

- Lactating → include milk
 - Dry → ignore milk
-

Multi-Signal Intelligence

Milk (if available)

Feed intake

Activity level

Temperature

Behavior

Voice (optional)

Simulation Engine (USP)

User modifies input → system recalculates risk

5. Target Users

- Dairy farmers
 - Livestock managers
 - Rural agricultural communities
-

6. Core Features

1. Animal Profile System

Animal ID
Breed
Age
State (lactating / dry)
Baseline behavior

2. Daily Tracking System

Milk (optional)
Feed intake
Activity
Temperature

3. Risk Engine

Hybrid system:

- Rule-based logic
 - ML-based estimation
-

4. Explainability

Reason:
- Milk drop
- Low activity
- High temperature

5. Simulation Engine

Before: 70
After: 52

6. Trend Tracking

Day-wise changes

Milk trends

Activity trends

7. System Flow

User Input



Validation



Feature Engineering



Context Check (state)



Risk Engine



Explainability



Response



Simulation (optional)

8. Input Specification

Milk: number

Feed: low/medium/high

Activity: low/medium/high

Temperature: °C

State: lactating/dry

9. Output

Risk Score: 72

Risk Level: HIGH

Reasons:

- Feed low
- Activity low

Suggested Action:

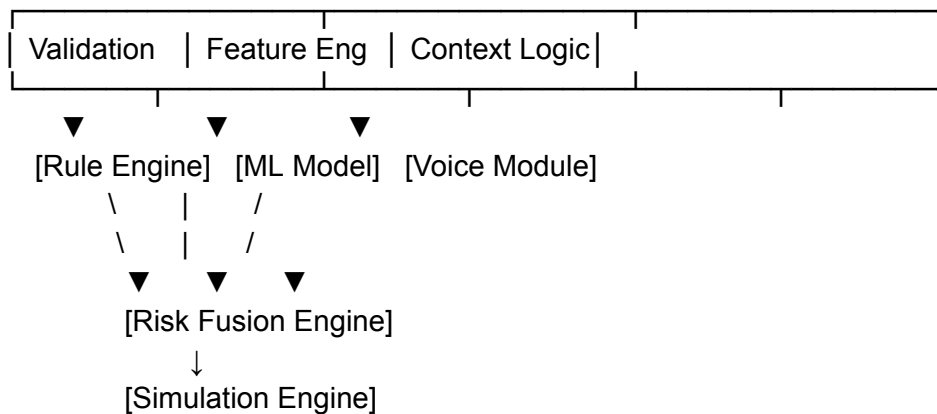
- Improve diet
 - Monitor closely
-

10. 🏗️ Architecture

Frontend (Farmer UI)



Backend (FastAPI)



11. ⚙️ Tech Stack

- Frontend: Next.js
 - Backend: FastAPI
 - ML: scikit-learn
 - Audio (optional): Librosa
 - Storage: SQLite
-

12. 📊 Success Metrics

- Accurate risk detection logic
- Smooth simulation
- Clear explainability

- Farmer usability
-

13. Limitations

- No clinical diagnosis
 - Manual data input
 - Limited dataset
-

14. Future Scope

- IoT sensor integration
 - Real-time monitoring
 - Multi-animal analytics
 - Mobile app
-

15. Final Positioning

“LHIS transforms livestock health management from reactive observation to proactive, data-driven decision-making.”
